

Adaptive Self-Consistency for Long-Form Factuality on Long-Tail Facts

Aggregation, metric sensitivity, and audited budget routing

Daniil Moskovskiy Rafael Grigoryan Evgenii Shuranov
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AIRI ■ Skoltech ■ MSU ■ ITMO University ■ HSE

KONVENS 2026

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Adaptive Self-Consistency

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Key message: aggregation can improve the executed factuality score, but the metric and budget evaluation require careful interpretation. Transition: first define the empirical questions and the audited evaluation.

Fluent long answers can hide many factual failures

The setting

- Entity-centric questions require **long, coherent answers**.
- Each answer contains many independently checkable claims.
- Rare entities make parametric recall less reliable.

A convincing answer
many atomic claims

Unsupported claims
remain persuasive

long tail

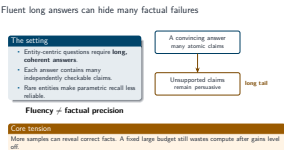
Core tension

More samples can reveal correct facts. A fixed large budget still wastes compute after gains level off.

Adaptive Self-Consistency

Fluent long answers can hide many factual failures

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Key message: the paper treats factuality as a claim-level problem and compute as a variable resource.
Transition: this motivates three empirical questions rather than a new evaluator or benchmark.

Three questions organize the study

1

Does sampling help?
Across head, torso, and tail entities.

2

How should candidates combine?
Synthesis versus voting and selection.

3

Can compute be routed?
Preserve factuality while reducing samples.

Empirical thesis
LLMAgg improves the executed FactOWL score; budget conclusions depend on the metric, and adaptive routing remains exploratory.

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└ Three questions organize the study

Key message: keep the talk focused on these three questions. Transition: answer them with a controlled multilingual, multi-domain evaluation.

Three questions organize the study

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Across head, torso, and tail entities.

2
How should candidates combine?
Synthesis versus voting and selection.

3
Can compute be routed?
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Empirical thesis
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Evaluation: LLM-judged FactOWL score across tiers

RiDiC benchmark

- 3 domains: rivers, cars, disasters.
- 2 languages: English and Chinese.
- Wikipedia pageview tiers: **head**, torso, **tail**.
- Public pool: up to 99 completions per input.

Reported score ($\gamma = 10$)

$$p(y) = \frac{1}{n} \sum_{i=1}^n 1[f_i \text{ supported}],$$
$$S_{10}(y) = p(y) \begin{cases} 1, & n > 10, \\ \exp(1 - 10/n), & n \leq 10. \end{cases}$$

Atomic facts and support labels are LLM-judged against the supplied Wikipedia page. For $n = 0$, $S_{10}(y) = 0$.

Lang.	Domain	Tier	RiDiC entity example
EN	cars	tail	Jiotto Caspita
EN	rivers	head	Blue Nile
EN	disasters	torso	Super Outbreak
ZH	cars	tail	劳斯莱斯 Silver Wraith
ZH	disasters	torso	2024 年花莲地震
ZH	rivers	head	伏尔塔瓦河

Adaptive Self-Consistency

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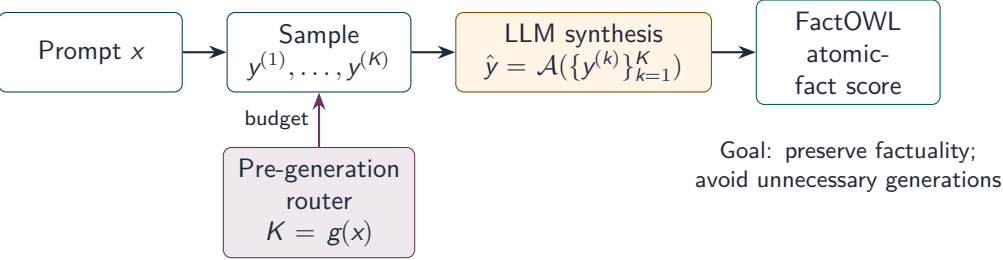
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Same generator: Llama 3.1 8B Instruct

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ZH	cars	tail	劳斯莱斯 Silver Wraith
ZH	disasters	torso	2024 年花莲地震
ZH	rivers	head	伏尔塔瓦河

Key message: all headline tables use the executed gamma-10 score, not plain factual precision. Fact extraction and support verification are model-judged proxies grounded in the supplied page. Transition: now show where sampling and synthesis enter the pipeline.

Method: sample, synthesize, score, then route the budget

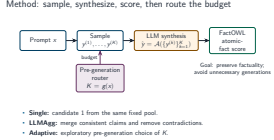


- **Single:** candidate 1 from the same fixed pool.
- **LLMAgg:** merge consistent claims and remove contradictions.
- **Adaptive:** exploratory pre-generation choice of K .

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Method: sample, synthesize, score, then route the budget



Key message: the intervention is content-level synthesis plus a separate pre-generation budget policy.
Transition: why synthesis matters becomes clearer when candidate answers disagree.

Synthesis has two distinct regimes

Complementary evidence

- Correct claims distributed across candidates.
- Agreement helps recover a shared core.
- Synthesis can exceed every input response.

Correlated error

- Candidates repeat related false claims.
- Agreement is not evidence of truth.
- Aggregation can amplify the wrong signal.

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└ Synthesis has two distinct regimes

Key message: synthesis helps only when the pool contains recoverable complementary evidence. Correlated errors create the opposite regime, so agreement alone is not a safety guarantee. Transition: compare the mechanisms explicitly.

Synthesis has two distinct regimes

Complementary evidence

- Correct claims distributed across candidates.
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- Synthesis can exceed every input response.

Correlated error

- Candidates repeat related false claims.
- Agreement is not evidence of truth.
- Aggregation can amplify the wrong signal.

Selection inherits one response; LLMAgg constructs a new response from the pool.

Baselines separate agreement, selection, and synthesis

Condition	What it does	Deployable?
Single	First completion from the same fixed pool	yes
MC	Majority co-occurrence over responses	yes
MF	Majority voting over normalized atomic facts	yes
Best-of- K logprob	Same selector on the matched $K = 20$ pool	yes
Best sampled	Oracle picks factuality-best candidate after scoring	no
LLMAgg	LLM synthesizes a new response from the pool	yes

Fair comparison

Single is candidate 1 of the shared pool. Matched- K methods use the same first 20 candidates; full-budget results use all 99.

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Adaptive Self-Consistency

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Fair comparison

Single is candidate 1 of the shared pool. Matched- K methods use the same first 20 candidates; full-budget results use all 99.

Key message: the paper does not compare one vague baseline against one vague method; it isolates response agreement, fact voting, logprob selection, oracle selection, and content synthesis. Transition: the main table shows a consistent ordering.

At matched $K = 20$, synthesis leads selection baselines

	Single	Best-of- K	ReASC	LLMAgg@20
English macro	0.281	0.304	0.290	0.577
Chinese macro	0.324	0.375	0.331	0.564

Across six language-domain cells

- Same first-20 candidate pool.
- LLMAgg leads both selection policies.
- Gains are broad, not driven by one domain.

Interpretation

These are executed S_{10} scores; they are not plain supported-fact precision.

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Adaptive Self-Consistency

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These are executed S_{10} scores; they are not plain supported-fact precision.

Key message: this is a strictly compute-matched comparison over the same first 20 candidates. LLMAgg reaches 0.577 in English and 0.564 in Chinese, ahead of the first candidate, log-probability selection, and corrected ReASC. Values use the executed gamma-10 score. Transition: quantify how the apparent budget threshold changes with the metric.

Operational K thresholds depend on the metric

Executed S_{10} : K_{95}/K_{99}

	EN	ZH
Cars	4/8	8/16
Disasters	8/20	8/16
Rivers	8/16	8/16

Raw support $p(y)$: K_{95}/K_{99}

	EN	ZH
Cars	1/1	1/8
Disasters	1/1	2/4
Rivers	1/1	4/16

Adaptive Self-Consistency

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Executed S_{10} : K_{95}/K_{99}		
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Rivers	1/1	4/16

K_{95} : earliest tested $K \in \{1, 2, 4, 8, 16, 20\}$ reaching 95% of the $K = 20$ score. No formal K_{99} is estimated.

Key message: the earlier claim of saturation around $K=8-16$ was metric-dependent. Under the executed gamma-10 score those are useful operational thresholds; under plain supported-fact precision, English reaches the K20 level at K1. Transition: popularity remains a separate structural axis.

Aggregation improves tail entities, but the head-to-tail gap remains

Lang.	Tier	Single macro S_{10}	LLMAgg@20 macro S_{10}
EN	head	0.445	0.828
EN	torso	0.384	0.762
EN	tail	0.243	0.513
ZH	head	0.480	0.757
ZH	torso	0.396	0.675
ZH	tail	0.275	0.501

Stable pattern

Head > torso > tail remains visible after improved inference.

Implication

Aggregation raises executed score broadly; long-tail difficulty remains.

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Adaptive Self-Consistency

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ZH	tail	0.275	0.501

Stable pattern
Head > torso > tail remains visible after improved inference.

Implication
Aggregation raises executed score broadly; long-tail difficulty remains.
Single is candidate 1 from the shared pool; values are macro averages over three domains.

Key message: on the executed gamma-10 score, LLMAgg improves every popularity tier but does not erase the head–tail gap. This is not a claim about an independently sampled Single. Transition: audit the adaptive-budget evidence.

The original $K = 99$ controller is not held-out evidence

Audit findings

- Eight baseline rows created per topic.
- Split performed row-wise, not by topic.
- Nearly every test topic appears in training.
- Raw score joins duplicate repeated titles.

Consequences

- Feature importance is not interpretable.
- Bootstrap rows are not independent.
- Reported $> 99\%$ retention is not a generalization result.
- Savings of 2–29% remain descriptive only.

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└ The original $K = 99$ controller is not held-out evidence

Key message: post-audit, the original K99 controller cannot support a held-out routing claim. The leakage comes from expanding each topic over baseline K values before a row-wise split. Transition: deduplicated K20-targeted recomputations give a more informative negative result.

The original $K = 99$ controller is not held-out evidence

Audit findings	Consequences
▪ Eight baseline rows created per topic. ▪ Split performed row-wise, not by topic. ▪ Nearly every test topic appears in training. ▪ Raw score joins duplicate repeated titles.	▪ Feature importance is not interpretable. ▪ Bootstrap rows are not independent. ▪ Reported $> 99\%$ retention is not a generalization result. ▪ Savings of 2–29% remain descriptive only.

The adaptive headline is withdrawn pending a topic-grouped rerun.

Deduplicated $K = 20$ reruns save compute but miss targets

Observed range

- Average K : **6.1–8.9**.
- Generation savings: **56–70%**.
- Score retention: **0.85–0.88**.
- Target coverage: **0.40–0.52**.

Why a range?

- Repeated titles have conflicting saved scores.
- Keep-first versus keep-last changes exact cells.
- Zero-score baselines make ratios undefined.
- The negative conclusion is unchanged.

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Adaptive Self-Consistency

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Compute fails, but the requested 0.95/0.99 retention is not achieved.

Key message: after enforcing unique topics and one-to-one score joins, K20-targeted reruns use roughly six to nine samples and save 56–70 percent. Exact cells vary with duplicate resolution, but retention stays around 0.85–0.88 and only 40–52 percent of nonzero-baseline examples meet target. Transition: separate what remains supported from what is still open.

Adaptive routing remains an open problem

Supported after audit

- Deduplicated reruns reduce average K .
- Some topics benefit substantially beyond $K = 20$.
- Example: Debby, 0.000 $\rightarrow$ 0.622 from $K = 20 \rightarrow 99$.

Not supported

- Reliable 0.95/0.99 retention.
- Generalization of the $K = 99$ policy.
- Detection of aggregation failures.
- A universal saturation budget.

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Adaptive Self-Consistency

└ Adaptive routing remains an open problem

Key message: the current evidence supports only an exploratory routing result. Large per-topic gains beyond K20 exist, but the tested features do not reliably identify them. A valid rerun must split by topic, enforce unique score keys, and define zero-baseline behavior before evaluation. Transition: close with the claims that survive the audit.

Adaptive routing remains an open problem

Supported after audit

- Deduplicated reruns reduce average K .
- Some topics benefit substantially beyond $K = 20$.
- Example: Debby, 0.000 $\rightarrow$ 0.622 from $K = 20 \rightarrow 99$.

Per-topic curves are non-monotonic.

Not supported

- Reliable 0.95/0.99 retention.
- Generalization of the $K = 99$ policy.
- Detection of aggregation failures.
- A universal saturation budget.

Use grouped splits and explicit zero handling.

Three takeaways

① Synthesis leads on the executed S_{10} score.

Single is candidate 1 from the same fixed pool.

② Budget thresholds are metric-dependent.

No formal or universal K_{sat} is established.

③ Adaptive routing is not yet validated.

Deduplicated K20 reruns save compute but miss retention targets.

Answer to the opening tension

Aggregation can help; efficient routing requires leakage-free evaluation.

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- Synthesis leads on the executed S_{10} score.
Single is candidate 1 from the same fixed pool.
- Budget thresholds are metric-dependent.
No formal or universal K_{sat} is established.
- Adaptive routing is not yet validated.
Deduplicated K20 reruns save compute but miss retention targets.

Answer to the opening tension

Aggregation can help; efficient routing requires leakage-free evaluation.

Key message: distinguish the executed score from raw factual precision, treat K thresholds as operational and metric-specific, and do not present the original adaptive controller as held-out evidence. Transition: references, then questions.

References

[D. Moskovskiy et al.](#)
Adaptive Self-Consistency for Long-Form Factuality on Long-Tail Facts. 2026.

[P. Braslavski et al.](#)
The Chronicles of RiDiC: Generating Datasets with Controlled Popularity Distribution for Long-form Factuality Evaluation. 2026.

[X. Wang et al.](#)
Self-Consistency Improves Chain of Thought Reasoning in Language Models. ICLR, 2023.

[J. Kim et al.](#)
Reliability-Aware Adaptive Self-Consistency for Efficient Sampling in LLM Reasoning. ACL Findings, 2026.

[A. Sakhovskiy et al.](#)
FactOWL: A Cost-Efficient Tool for Long-Form Factuality Evaluation. SIGIR, 2026.

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References

Key message: the primary paper builds on self-consistency and fact-level factuality evaluation, but studies their interaction under long-tail and budget constraints. Transition: thank the audience and invite questions.

References

[D. Moskovskiy et al.](#)
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Thank you

Questions?

Daniil Moskovskiy | A.Panchenko@skol.tech

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└ Thank you

Thank you

Questions?

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Close: aggregation improves the executed score, while metric sensitivity and adaptive-budget validation remain open. Invite questions about the audit, evaluator reliability, and per-topic non-monotonicity.

Backup slides

Audited tables, definitions, and case traces

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Backup slides

Backup slides

Audited tables, definitions, and case traces

Use these slides for questions about exact per-cell values, policy definitions, dataset filtering, or qualitative behavior.

Matched- $K = 20$ comparison by language and domain

Lang.	Domain	Single	LLMAgg@20	Gain
EN	rivers	0.211	0.463	+0.252
EN	cars	0.344	0.655	+0.311
EN	disasters	0.287	0.612	+0.325
ZH	rivers	0.269	0.442	+0.173
ZH	cars	0.349	0.633	+0.285
ZH	disasters	0.354	0.616	+0.262

Reading the table

Values use executed S_{10} on the same matched subset. Single is candidate 1 from the first-20 pool.

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Adaptive Self-Consistency

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Reading the table
Values use executed S_{10} on the same matched subset. Single is candidate 1 from the first-20 pool.

Use this slide to answer questions about individual cells behind the macro averages.

Supplementary baselines at matched $K = 20$

Lang.	Domain	Single	Best-of- K logprob	ReASC	LLMAgg@20
EN	cars	0.344	0.366	0.354	0.655
EN	disasters	0.287	0.324	0.305	0.612
EN	rivers	0.211	0.221	0.210	0.463
ZH	cars	0.349	0.407	0.349	0.633
ZH	disasters	0.354	0.417	0.368	0.616
ZH	rivers	0.269	0.302	0.277	0.442

Audit note

ReASC is the corrected post-hoc baseline: completion-level FactOWL scores replace the earlier topic-level join.

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Adaptive Self-Consistency

Supplementary baselines at matched $K = 20$

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ZH	rivers	0.269	0.302	0.277	0.442

Audit note
ReASC is the corrected post-hoc baseline: completion-level FactOWL scores replace the earlier topic-level join.
Best-of- K logprob and ReASC are compared with LLMAgg at the same $K = 20$ budget.

Use this slide when asked whether synthesis still beats stronger selection signals at a matched budget.
ReASC is reported after correcting the topic-level join bug described in the paper.

Required protocol for a valid adaptive rerun

Well-defined quantities

$$\begin{aligned} \text{compute savings} &= 1 - \frac{\sum_i K_i^{\text{adaptive}}}{N K_{\text{full}}} \\ \text{aggregate retention} &= \frac{\sum_i S_{10}(\hat{y}_i^{\text{adaptive}})}{\sum_i S_{10}(\hat{y}_i^{\text{full}})} \\ \text{coverage} &= \Pr[S_i^{\text{adaptive}} \geq \tau S_i^{\text{full}} \mid S_i^{\text{full}} > 0] \end{aligned}$$

Leakage controls

- Group split by language, domain, topic.
- One row per (topic, K) score.
- Report zero-score baselines separately.
- Bootstrap unique topics, not fold rows.
- Freeze metric before model selection.

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Adaptive Self-Consistency

Required protocol for a valid adaptive rerun

Use this slide to explain the minimum protocol required before making a new adaptive-routing claim. Aggregate retention avoids averaging unstable per-example ratios, while coverage explicitly conditions on a nonzero baseline.

Well-defined quantities
compute savings $\coloneqq 1 - \frac{\sum_i K_i^{\text{adaptive}}}{N K_{\text{full}}}$
aggregate retention $\coloneqq \frac{\sum_i S_{10}(\hat{y}_i^{\text{adaptive}})}{\sum_i S_{10}(\hat{y}_i^{\text{full}})}$
coverage $\coloneqq \Pr[S_i^{\text{adaptive}} \geq \tau S_i^{\text{full}} \mid S_i^{\text{full}} > 0]$

Leakage controls

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- One row per (topic, K) score.
- Report zero-score baselines separately.
- Bootstrap unique topics, not fold rows.
- Freeze metric before model selection.

Generation-count savings still exclude aggregation-to-len cost; report both when deployment cost matters.

Dataset filtering and FactOWL details

Saved evaluation rows

	rivers	cars	disasters
English	1000	997	1000
Chinese	723	400	723

Executed FactOWL score

$$F(y) = \{f_1, \dots, f_n\}$$
$$p(y) = \frac{1}{n} \sum_i 1[f_i \text{ supported}]$$
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Verification uses the supplied page; fact extraction and support labels are LLM-generated proxies. For $n = 0$, $S_{10}(y) = 0$.

Adaptive Self-Consistency

Dataset filtering and FactOWL details

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Dataset filtering and FactOWL details

Saved evaluation rows			
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Counts are saved evaluation rows, not a fresh recount of cached nonempty contexts.

Executed FactOWL score	
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$p(y) = \frac{1}{n} \sum_i 1[f_i \text{ supported}]$	
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Verification uses the supplied page; fact extraction and support labels are LLM-generated proxies. For $n = 0$, $S_{10}(y) = 0$.	

Use this slide when asked whether language sample counts match or why short answers receive lower scores. The executed score includes a gamma-10 penalty; raw supported-fact precision is retained separately as init score.

Audited case: Debby collapses at $K = 20$, then recovers

Executed S_{10} curve

K	score
1	0.221
2	0.428
4	0.500
8	0.118
16	0.083
20	0.000
32	0.125
64	0.286
99	0.622

What the audit established

- Single is candidate 1 in the pool.
- Published 0.548 pooled facts from all 99 candidates.
- Correct Single score: 0.221; raw support: 3/5.
- No first-20 candidate identifies 2024.
- Wrong years differ, but errors are correlated.

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Adaptive Self-Consistency

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- Wrong years differ, but errors are correlated.

Evidence of correlated error in this setup, not a causal limitation of Llama.

Use this slide for the Debby question. The earlier 0.548 value was not a Single score but pooled support over all 99 candidate facts. The first 20 candidates contain different wrong years and no correct 2024 identification, so K20 synthesizes a correlated error. K99 recovers in metric score, although its text still does not correctly identify the 2024 storm.